

Course Map

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1. Overview

Key point: The Course map ties every Note together. It shows where each idea sits in an ML project, how ideas connect, what to read first, and which algorithm to choose.

The Notes each teach one lesson. The **Course map** (G-2160) shows how those lessons fit together, in four views:

View	Question it answers
Pipeline map (G-2161)	Where does this idea sit in a real ML project?
Concept map (G-2162)	How is this idea connected to the others?
Learning path (G-2163)	Which Notes should I read first?
Algorithm chooser (G-2164)	Which algorithm suits my problem?

Each idea on the map is a **Concept** (G-2165). Concepts already taught in a written Note are **confirmed**; the others are **draft** (G-2167), shown faint, and are checked against their source when their Note is written. So far, 335 of 335 Concepts are confirmed.

Every Note starts with a *Where this fits* box: a small Pipeline map with that Note's steps highlighted, plus what it builds on and what it leads to.

Extra: This map is generated from one data file, `course_map/concepts.yaml`. An interactive version, where you can zoom, filter by step and click a Concept to see its links and Notes, runs with `python course_map/app.py`.

2. The Pipeline map

Key point: Every ML project moves through the same steps, from framing the problem to monitoring the deployed model. Every Concept belongs to one step.

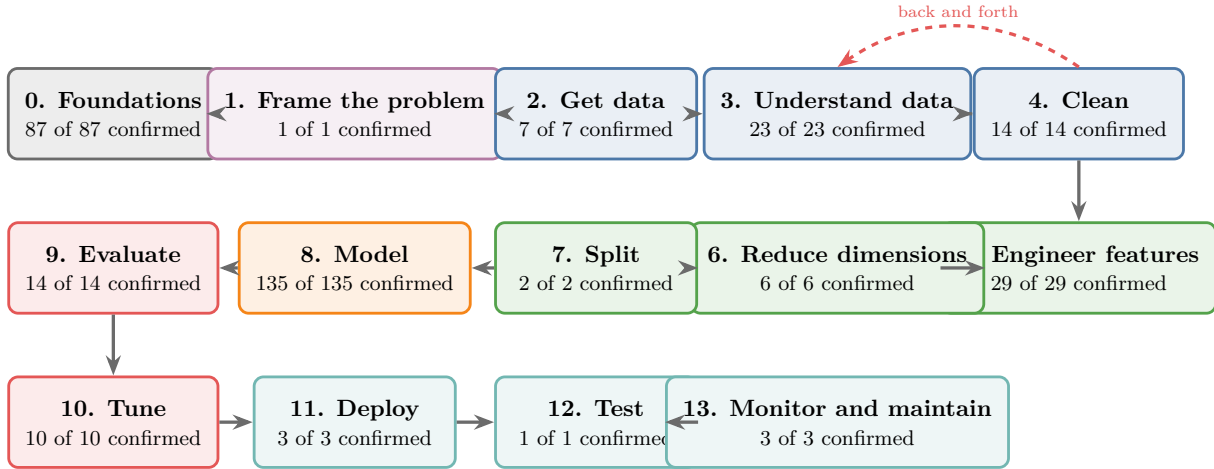


Figure 1: The Pipeline map: 14 steps of an ML project

Figure 1 shows the 14 steps. They follow the ML development life cycle, or **MLDLC** (G-1240):

1. frame the problem;
2. get the data and understand it;
3. clean the data;
4. engineer features;
5. train and evaluate models;
6. tune them;
7. deploy, test and keep the model healthy.

Steps 3 (Understand data) and 4 (Clean) loop: exploring reveals what needs cleaning, and cleaning changes the data, so we explore again.

2.1 Step 0: Foundations

Concept	Taught in	Status
Machine learning	Note 1 , Note 2	confirmed
Data mining	Note 1 , Note 8	confirmed
Artificial intelligence	Note 2	confirmed
Symbolic AI and expert systems	Note 2	confirmed
Deep learning	Note 2	confirmed
Neural networks	Note 2 , Note 1002	confirmed
Features	Note 2 , Note 11	confirmed
Supervised learning	Note 3	confirmed
Regression problems	Note 3	confirmed
Classification problems	Note 3	confirmed

Concept	Taught in	Status
Unsupervised learning	Note 3	confirmed
Semi-supervised learning	Note 3	confirmed
Reinforcement learning	Note 3	confirmed
Batch (offline) learning	Note 4	confirmed
Online learning	Note 5	confirmed
Out-of-core learning	Note 5	confirmed
Instance-based learning	Note 6	confirmed
Model-based learning	Note 6	confirmed
Applications of ML	Note 8	confirmed
ML development life cycle	Note 9 , Note 13	confirmed
Tensors	Note 11	confirmed
Setup: conda, Jupyter and Colab	Note 12	confirmed
Chi-square tests	Note 571 , Note 29	confirmed
Vector magnitude, distance and scalar operations	Note 39 , Note 361	confirmed
Normal distribution	Note 42 , Note 90 , Note 240 , Note 250 , Note 630	confirmed
Eigenvectors and eigenvalues	Note 48 , Note 530	confirmed
Dot product	Note 48 , Note 362 , Note 520	confirmed
Linear transformations and matrices	Note 48 , Note 500	confirmed
Derivatives of one variable	Note 51 , Note 600	confirmed
Equation of a hyperplane	Note 53 , Note 70 , Note 363	confirmed
Partial derivatives and gradients	Note 57 , Note 601	confirmed
Conditional probability	Note 82 , Note 341	confirmed
Independent and mutually exclusive events	Note 83 , Note 84	confirmed
Bayes' theorem	Note 85 , Note 86 , Note 341	confirmed
Bernoulli and binomial distributions	Note 102 , Note 241 , Note 270	confirmed
Hessian and multivariate Taylor	Note 126 , Note 603	confirmed
Taylor series	Note 126 , Note 600 , Note 603	confirmed
Probability distributions	Note 210 , Note 240	confirmed
Inferential statistics	Note 210 , Note 220	confirmed
Population, sample, parameter and statistic	Note 220	confirmed
Random variables	Note 240	confirmed
Probability mass function (PMF)	Note 241	confirmed
Uniform distribution	Note 241 , Note 261	confirmed

Concept	Taught in	Status
Log-normal distribution	Note 242 , Note 261	confirmed
Poisson distribution	Note 242 , Note 560	confirmed
Standard normal and the z-table	Note 251	confirmed
Pareto distribution and power laws	Note 262	confirmed
Sampling distribution and standard error	Note 271	confirmed
Central limit theorem	Note 271 , Note 272	confirmed
Confidence intervals	Note 280 , Note 281 , Note 282	confirmed
Student's t-distribution	Note 282	confirmed
Hypothesis testing: null and alternative	Note 290 , Note 291	confirmed
Z-test and rejection regions	Note 291	confirmed
Type I and II errors, power, tails	Note 292	confirmed
P-values	Note 300	confirmed
T-tests: one-sample, two-sample, paired	Note 301 , Note 302 , Note 570	confirmed
Events and sample spaces	Note 330	confirmed
Empirical vs theoretical probability, probability rules	Note 331 , Note 340	confirmed
Expected value and variance of a random variable	Note 332	confirmed
Venn diagrams and contingency tables	Note 340	confirmed
Joint and marginal probability	Note 341	confirmed
Linear algebra roadmap	Note 350	confirmed
Vectors and feature vectors	Note 360	confirmed
Role of mathematics in ML	Note 440	confirmed
Linear combinations, span and basis	Note 490	confirmed
Matrix multiplication as composition	Note 510	confirmed
Determinant	Note 530	confirmed
Choosing a hypothesis test	Note 570	confirmed
One-sample proportion test	Note 570	confirmed
One-way ANOVA	Note 572	confirmed
How to learn the maths for ML	Note 580	confirmed
Jacobian and matrix gradients	Note 602	confirmed

Concept	Taught in	Status
Singular value decomposition	Note 610 , Note 611	confirmed
Lagrange multipliers, KKT and duality	Note 620	confirmed
Convex sets and convex optimisation	Note 621	confirmed
Linear and quadratic programming	Note 622	confirmed
Likelihood	Note 630	confirmed
Maximum likelihood estimation (MLE)	Note 631 , Note 632 , Note 633	confirmed
Exponential distribution	Note 632	confirmed
Multivariate normal distribution	Note 640	confirmed
What deep learning is	Note 1001 , Note 1002	confirmed
Representation learning	Note 1002	confirmed
History of deep learning	Note 1003	confirmed
Universal approximation theorem	Note 1003 , Note 1009	confirmed
Memoization	Note 1019	confirmed
Exponentially weighted moving average (EWMA)	Note 1033	confirmed
Sequential data	Note 1055	confirmed

2.2 Step 1: Frame the problem

Concept	Taught in	Status
Framing an ML problem	Note 9 , Note 14	confirmed

2.3 Step 2: Get data

Concept	Taught in	Status
Labelled data	Note 3 , Note 7	confirmed
Enough data	Note 7	confirmed
Sampling noise and bias	Note 7	confirmed
APIs	Note 7 , Note 9 , Note 17	confirmed
Web scraping	Note 7 , Note 9 , Note 18	confirmed
CSV files	Note 9 , Note 13 , Note 15	confirmed
JSON and SQL data	Note 16	confirmed

2.4 Step 3: Understand data

Concept	Taught in	Status
Exploratory data analysis	Note 9 , Note 13 , Note 19	confirmed
Univariate analysis	Note 9 , Note 20	confirmed
Bivariate and multivariate analysis	Note 9 , Note 21	confirmed
Imbalanced data	Note 9 , Note 133	confirmed
Variance	Note 19 , Note 47 , Note 222	confirmed
Correlation	Note 19 , Note 21 , Note 231	confirmed
Descriptive statistics	Note 19 , Note 210 , Note 221 , Note 222 , Note 223 , Note 230	confirmed
Probability density function (PDF)	Note 20 , Note 90 , Note 242	confirmed
Kernel density estimation (KDE)	Note 20 , Note 243 , Note 253	confirmed
Skewness	Note 20 , Note 252	confirmed
Kurtosis and moments	Note 22 , Note 260	confirmed
Pandas Profiling	Note 22	confirmed
Q-Q plot	Note 30 , Note 260	confirmed
Covariance and covariance matrix	Note 48 , Note 231	confirmed
Discrete and continuous data	Note 220	confirmed
Measures of central tendency	Note 221	confirmed
Bessel's correction	Note 222	confirmed
Frequency tables	Note 223	confirmed
Percentiles, quartiles and box plots	Note 230	confirmed
Correlation and causation	Note 231	confirmed
Cumulative distribution function (CDF)	Note 241 , Note 242 , Note 253	confirmed
Density estimation	Note 243	confirmed
Correlation significance test	Note 570	confirmed

2.5 Step 4: Clean

Concept	Taught in	Status
Poor-quality data	Note 7 , Note 9	confirmed
Missing values	Note 7 , Note 9 , Note 23 , Note 35 , Note 36 , Note 37 , Note 38 , Note 123	confirmed
Outliers	Note 7 , Note 9 , Note 20 , Note 23 , Note 41	confirmed
Simple imputation (mean, median, mode, constant)	Note 23 , Note 36 , Note 37	confirmed

Concept	Taught in	Status
Complete case analysis	Note 35	confirmed
Missing indicator	Note 38	confirmed
Random sample imputation	Note 38	confirmed
KNN imputer	Note 39	confirmed
Iterative imputation (MICE)	Note 40	confirmed
Trimming outliers	Note 41 , Note 42 , Note 43 , Note 44	confirmed
Capping (winsorization)	Note 41 , Note 42 , Note 43 , Note 44	confirmed
Z-score outlier method	Note 41 , Note 42 , Note 251	confirmed
IQR outlier method	Note 41 , Note 43	confirmed
Percentile outlier method	Note 41 , Note 44	confirmed

2.6 Step 5: Engineer features

Concept	Taught in	Status
Feature construction and splitting	Note 3 , Note 7 , Note 9 , Note 23 , Note 45	confirmed
Feature scaling	Note 6 , Note 9 , Note 13 , Note 23 , Note 24 , Note 1023	confirmed
Feature engineering	Note 7 , Note 9 , Note 23	confirmed
Feature selection	Note 9 , Note 13 , Note 23 , Note 46	confirmed
Standardization	Note 9 , Note 13 , Note 24 , Note 1023 , Note 1031	confirmed
One-hot encoding	Note 11 , Note 23 , Note 27	confirmed
ML pipelines	Note 13 , Note 29 , Note 38	confirmed
Encoding categorical data	Note 23 , Note 26	confirmed
Binning and binarization	Note 23 , Note 32	confirmed
Feature transformation	Note 23	confirmed
Normalization	Note 25	confirmed
Ordinal and label encoding	Note 26	confirmed
Column transformer	Note 28	confirmed
Function transformer	Note 30	confirmed
Power transformer	Note 31	confirmed
Mixed variables	Note 33	confirmed
Date and time features	Note 34	confirmed
Feature importance	Note 99 , Note 100 , Note 114	confirmed
Permutation importance	Note 114	confirmed
Random under- and oversampling	Note 133	confirmed
SMOTE	Note 133	confirmed

Concept	Taught in	Status
Bag of words	Note 360	confirmed
Scaling inputs for neural networks	Note 1023	confirmed
Data augmentation	Note 1050	confirmed
Sequence padding	Note 1055 , Note 1057	confirmed
Tokenization and integer encoding of text	Note 1057	confirmed
Word embeddings	Note 1057	confirmed
Contextual embeddings	Note 1072	confirmed
Meaning as direction in embedding space	Note 1086	confirmed

2.7 Step 6: Reduce dimensions

Concept	Taught in	Status
Dimensionality reduction	Note 3 , Note 46	confirmed
PCA	Note 3 , Note 47 , Note 48 , Note 49	confirmed
Feature extraction	Note 23 , Note 46 , Note 47	confirmed
Curse of dimensionality	Note 46 , Note 91	confirmed
Low-rank approximation (truncated SVD)	Note 612	confirmed
Latent semantic analysis	Note 613	confirmed

2.8 Step 7: Split

Concept	Taught in	Status
Train-test split	Note 13	confirmed
Data leakage	Note 13 , Note 91	confirmed

2.9 Step 8: Model

Concept	Taught in	Status
Clustering	Note 3 , Note 128 , Note 129 , Note 130 , Note 131 , Note 132	confirmed
Anomaly detection	Note 3 , Note 132	confirmed
Association rule learning	Note 3	confirmed
Stochastic gradient descent	Note 5 , Note 59 , Note 1020	confirmed
K-nearest neighbours	Note 6 , Note 91	confirmed

Concept	Taught in	Status
Ensemble learning	Note 9 , Note 101	confirmed
Logistic regression	Note 13 , Note 70 , Note 71 , Note 72 , Note 73 , Note 75	confirmed
Multicollinearity	Note 27	confirmed
K-means	Note 32 , Note 128 , Note 129 , Note 130 , Note 641	confirmed
Simple linear regression	Note 50 , Note 51	confirmed
Best-fit line and squared error	Note 50 , Note 51	confirmed
Ordinary least squares (closed form)	Note 51	confirmed
Multiple linear regression	Note 53 , Note 54 , Note 55	confirmed
Normal equation	Note 54 , Note 55	confirmed
Assumptions of linear regression	Note 56	confirmed
Gradient descent	Note 57 , Note 1017 , Note 1020	confirmed
Convex and non-convex loss	Note 57 , Note 590 , Note 1017	confirmed
Batch gradient descent	Note 58 , Note 1020	confirmed
Mini-batch gradient descent	Note 60 , Note 1020	confirmed
Polynomial regression	Note 61	confirmed
Polynomial features	Note 61 , Note 80	confirmed
Regularisation	Note 63 , Note 1026	confirmed
Ridge regression	Note 63 , Note 64 , Note 65 , Note 66	confirmed
Lasso regression	Note 67 , Note 68	confirmed
Elastic Net	Note 69	confirmed
Perceptron trick	Note 70 , Note 71 , Note 1005	confirmed
Sigmoid function	Note 72 , Note 74 , Note 1018 , Note 1027	confirmed
Log loss (binary cross entropy)	Note 73 , Note 1014 , Note 633	confirmed
Softmax regression	Note 79	confirmed
Naive Bayes	Note 87 , Note 88 , Note 89 , Note 90	confirmed
Support vector machines	Note 92 , Note 93 , Note 94	confirmed
Hinge loss and soft margin	Note 94	confirmed
Kernel trick	Note 95 , Note 96	confirmed
Entropy, information gain and Gini	Note 97	confirmed
Decision trees	Note 97 , Note 98 , Note 100	confirmed
Regression trees	Note 99	confirmed
Boosting	Note 101 , Note 119	confirmed

Concept	Taught in	Status
Voting ensembles	Note 102 , Note 103 , Note 104	confirmed
Bagging	Note 105 , Note 106 , Note 107	confirmed
Random forest	Note 108 , Note 109 , Note 110 , Note 111 , Note 112 , Note 113 , Note 114	confirmed
AdaBoost	Note 115 , Note 116 , Note 117 , Note 118	confirmed
Gradient boosting	Note 120 , Note 121 , Note 122	confirmed
XGBoost	Note 123 , Note 124 , Note 125 , Note 126	confirmed
Stacking and blending	Note 127	confirmed
Hierarchical clustering	Note 131	confirmed
DBSCAN	Note 132	confirmed
Balanced random forest	Note 133	confirmed
Cost-sensitive learning	Note 133	confirmed
Cosine similarity	Note 362	confirmed
Moore-Penrose pseudo-inverse	Note 613	confirmed
Categorical and sparse categorical cross-entropy	Note 1012 , Note 1014 , Note 633	confirmed
MAP estimation	Note 633	confirmed
Gaussian mixture model (GMM)	Note 640 , Note 641	confirmed
Expectation maximization (EM)	Note 641	confirmed
Types of neural networks	Note 1003	confirmed
Multi-layer perceptron (MLP)	Note 1003 , Note 1009	confirmed
Perceptron	Note 1004	confirmed
Perceptron loss	Note 1006	confirmed
Problem with the perceptron (XOR)	Note 1007	confirmed
MLP notation and parameter count	Note 1008	confirmed
Forward propagation	Note 1010	confirmed
Keras workflow	Note 1011 , Note 1012 , Note 1013 , Note 1022 , Note 1025 , Note 1026	confirmed
ANN for classification	Note 1011 , Note 1012	confirmed
ANN for regression	Note 1013	confirmed
Loss functions in deep learning	Note 1014	confirmed
Huber loss	Note 1014	confirmed

Concept	Taught in	Status
Backpropagation	Note 1015 , Note 1016 , Note 1017 , Note 1019	confirmed
Vanishing gradient	Note 1018 , Note 1029 , Note 1030 , Note 1060	confirmed
Exploding gradient and gradient clipping	Note 1018 , Note 1029 , Note 1060	confirmed
ReLU	Note 1018 , Note 1027 , Note 1028	confirmed
Batch size in Keras	Note 1020	confirmed
Dropout	Note 1024 , Note 1025	confirmed
L1 and L2 regularisation in neural networks	Note 1026	confirmed
Activation functions	Note 1027	confirmed
Tanh	Note 1027	confirmed
Dying ReLU problem	Note 1028	confirmed
Leaky ReLU, PReLU, ELU and SELU	Note 1028	confirmed
Weight initialisation	Note 1029 , Note 1030	confirmed
Xavier and He initialisation	Note 1030	confirmed
Batch normalisation	Note 1031	confirmed
Covariate shift	Note 1031	confirmed
Optimizers in deep learning	Note 1032	confirmed
Local minima and saddle points	Note 1032 , Note 1034	confirmed
SGD with momentum	Note 1034	confirmed
Nesterov accelerated gradient (NAG)	Note 1035	confirmed
AdaGrad	Note 1036	confirmed
RMSProp	Note 1037	confirmed
Adam	Note 1038	confirmed
Convolutional neural network (CNN)	Note 1040 , Note 1041 , Note 1046	confirmed
Convolution operation and feature maps	Note 1042	confirmed
Padding and strides	Note 1043	confirmed
Pooling	Note 1044	confirmed
CNN architecture (LeNet-5)	Note 1045	confirmed
Backpropagation in a CNN	Note 1047 , Note 1048	confirmed
Image classification with a CNN (cats vs dogs)	Note 1049	confirmed
Pretrained models and ImageNet	Note 1051	confirmed

Concept	Taught in	Status
Transfer learning (feature extraction and fine-tuning)	Note 1053	confirmed
Keras functional API	Note 1054	confirmed
Skip connections	Note 1054	confirmed
Recurrent neural network (RNN)	Note 1055 , Note 1056 , Note 1057	confirmed
Parameter sharing across time steps	Note 1056 , Note 1059	confirmed
Types of RNN (many-to-one, one-to-many, many-to-many)	Note 1058	confirmed
Sequence-to-sequence (encoder-decoder)	Note 1058 , Note 1068	confirmed
Backpropagation through time (BPTT)	Note 1059	confirmed
Long-term dependency problem	Note 1060	confirmed
LSTM (long short-term memory)	Note 1061 , Note 1062 , Note 1063	confirmed
LSTM gates (forget, input, output) and cell state	Note 1062	confirmed
Next-word prediction with an LSTM	Note 1063	confirmed
GRU (gated recurrent unit)	Note 1064	confirmed
Deep (stacked) RNNs	Note 1065	confirmed
Bidirectional RNNs	Note 1066	confirmed
Large language models (LLMs)	Note 1067	confirmed
Teacher forcing	Note 1068 , Note 1081	confirmed
Attention mechanism	Note 1069	confirmed
Bahdanau (additive) attention	Note 1069 , Note 1070	confirmed
Luong (multiplicative) attention	Note 1070	confirmed
Transformer	Note 1071 , Note 1080 , Note 1083	confirmed
Self-attention (query, key, value)	Note 1072 , Note 1073 , Note 1075 , Note 1076	confirmed
Scaled dot-product attention	Note 1074	confirmed
Multi-head attention	Note 1077	confirmed
Positional encoding	Note 1078	confirmed
Layer normalisation	Note 1079	confirmed
Residual connections and add & norm	Note 1080	confirmed

Concept	Taught in	Status
Transformer encoder	Note 1080	confirmed
Masked self-attention	Note 1081	confirmed
Cross-attention	Note 1082	confirmed
Transformer decoder	Note 1083	confirmed
Transformer inference (autoregressive decoding, KV cache, beam search)	Note 1084	confirmed
The transformer end to end (capstone)	Note 1085	confirmed
Label smoothing	Note 1085	confirmed
Decoder-only GPT	Note 1087	confirmed
GELU activation	Note 1087	confirmed
Unembedding, logits, temperature and sampling	Note 1088	confirmed
MLP blocks as fact storage	Note 1089	confirmed
Superposition and nearly perpendicular directions	Note 1090	confirmed

2.10 Step 9: Evaluate

Concept	Taught in	Status
Overfitting	Note 7 , Note 61 , Note 91 , Note 1026	confirmed
Underfitting	Note 7 , Note 61 , Note 91	confirmed
Accuracy	Note 13 , Note 76 , Note 91	confirmed
Cross-validation	Note 29 , Note 30 , Note 91 , Note 103 , Note 104 , Note 112 , Note 118 , Note 127	confirmed
Regression metrics	Note 52	confirmed
Bias-variance trade-off	Note 62 , Note 109	confirmed
Confusion matrix	Note 76	confirmed
Precision, recall and F1	Note 77	confirmed
ROC curve and AUC	Note 78	confirmed
Decision surface and boundary	Note 91	confirmed
OOB score	Note 105 , Note 106 , Note 107 , Note 113	confirmed
Training curves (History)	Note 1011 , Note 1012 , Note 1013 , Note 1022	confirmed
Visualising what a CNN learns	Note 1052	confirmed
Logit lens	Note 1088	confirmed

Concept	Taught in	Status
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2.11 Step 10: Tune

Concept	Taught in	Status
Hyperparameter tuning	Note 9 , Note 81 , Note 91 , Note 98 , Note 111 , Note 118 , Note 1021 , Note 1039	confirmed
Grid and random search	Note 29 , Note 38 , Note 91 , Note 99 , Note 106 , Note 107 , Note 112 , Note 118	confirmed
Learning rate	Note 57 , Note 118 , Note 120 , Note 1017	confirmed
Elbow method and WCSS	Note 128 , Note 129	confirmed
Bayesian optimisation	Note 134	confirmed
Optuna	Note 134	confirmed
Improving a neural network	Note 1021	confirmed
Early stopping	Note 1021 , Note 1022	confirmed
Keras Tuner	Note 1039	confirmed
Learning-rate warm-up schedule	Note 1085	confirmed

2.12 Step 11: Deploy

Concept	Taught in	Status
Deployment	Note 4 , Note 7 , Note 9 , Note 13 , Note 29	confirmed
Software integration	Note 7	confirmed
Saving models with pickle	Note 9 , Note 13 , Note 29	confirmed

2.13 Step 12: Test

Concept	Taught in	Status
Beta and A/B testing	Note 9 , Note 292	confirmed

2.14 Step 13: Monitor and maintain

Concept	Taught in	Status
Model drift	Note 4 , Note 9	confirmed

Concept	Taught in	Status
Retraining	Note 4 , Note 5 , Note 9	confirmed
MLOps and cost	Note 7 , Note 9	confirmed

3. The Concept map

Key point: Concepts are joined by five kinds of Link. Following the Links shows why each idea exists and what it is for.

Every **Link** (G-2166) has one of five types:

Link	Meaning	Example
needs	must be understood first	logistic regression <i>needs</i> gradient descent
is a kind of	a special case	Ridge <i>is a kind of</i> regularisation
fixes	solves a problem	regularisation <i>fixes</i> overfitting
compared with	often confused or contrasted	bagging <i>compared with</i> boosting
used in	a tool used inside something else	feature scaling <i>used in</i> KNN

The full map has 335 Concepts, too many for one page, so each topic has its own mind map (Figures 2 to 19). Each map tells the topic's story from left to right and shows only its key Concepts and links; the grey number in each box is the Note that teaches it, and dashed boxes lead to other maps. To see every Concept and every link at once, open `course_map/concept_map_3d.html` in a browser: a 3D network you can rotate and zoom, with one colour per area (rebuilt by `python course_map/map_3d.py`). The interactive app (`python course_map/app.py`) lists each Concept's links with the Notes that teach them.

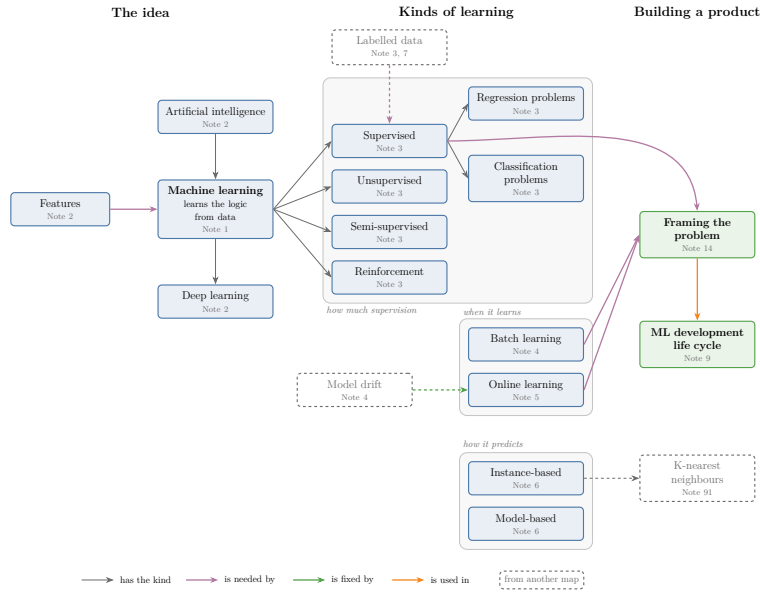


Figure 2: Concept map: Foundations and framing

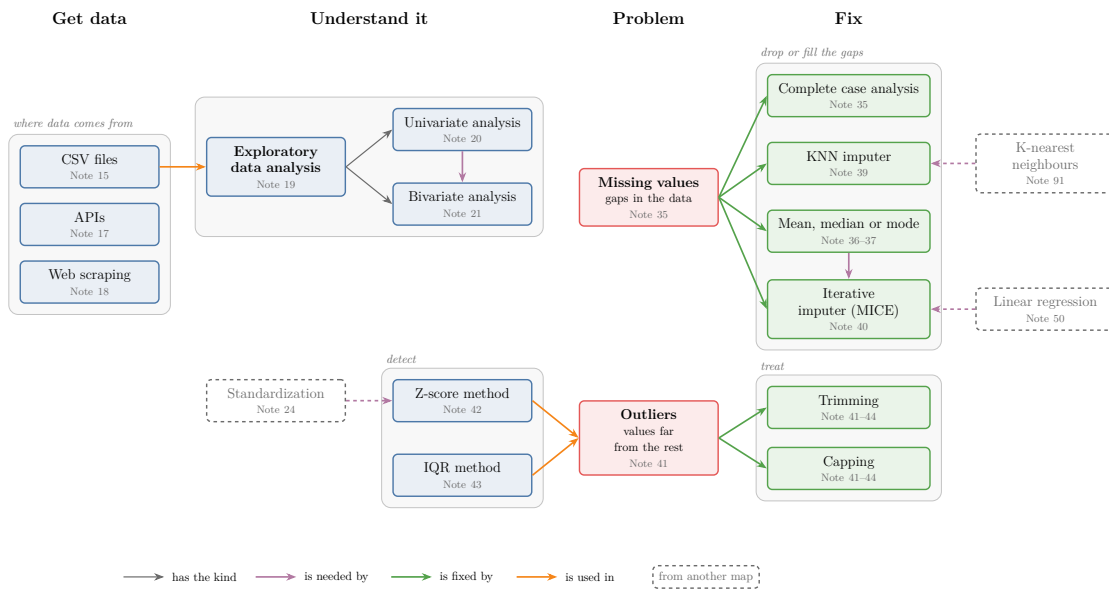


Figure 3: Concept map: Getting, understanding and cleaning data

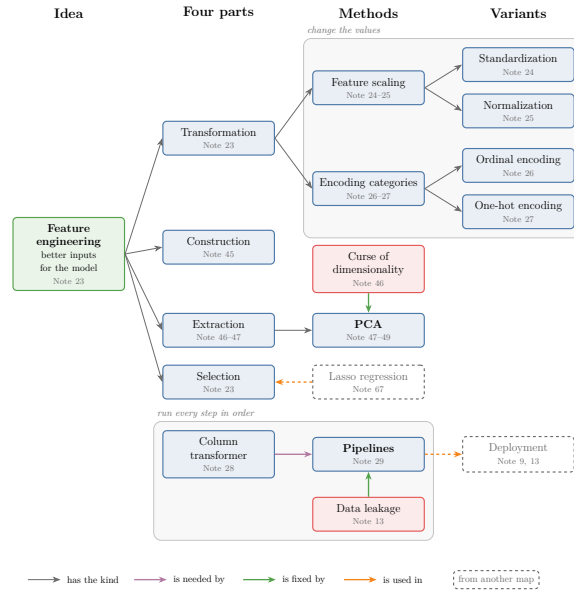


Figure 4: Concept map: Features, dimensions and splitting

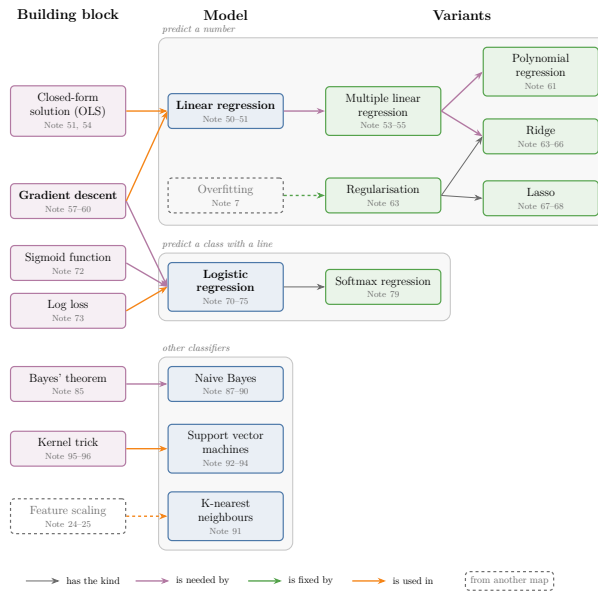


Figure 5: Concept map: Models: regression, classification and gradient descent

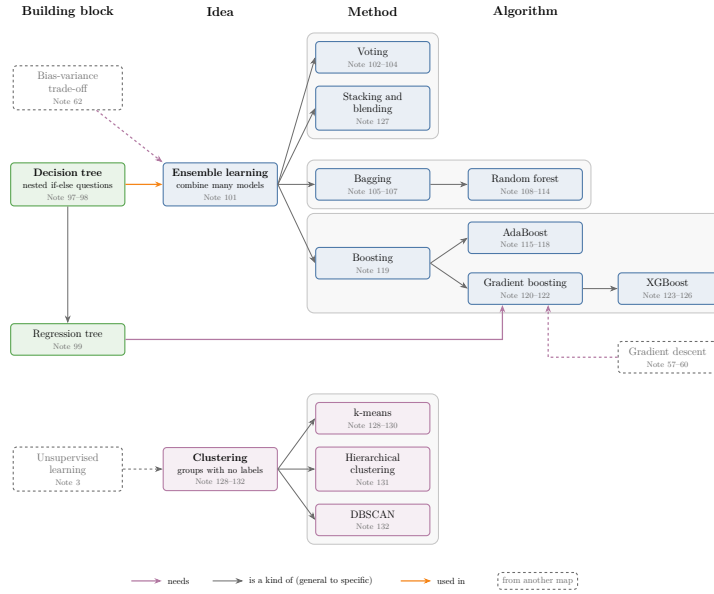


Figure 6: Concept map: Models: trees, ensembles and clustering

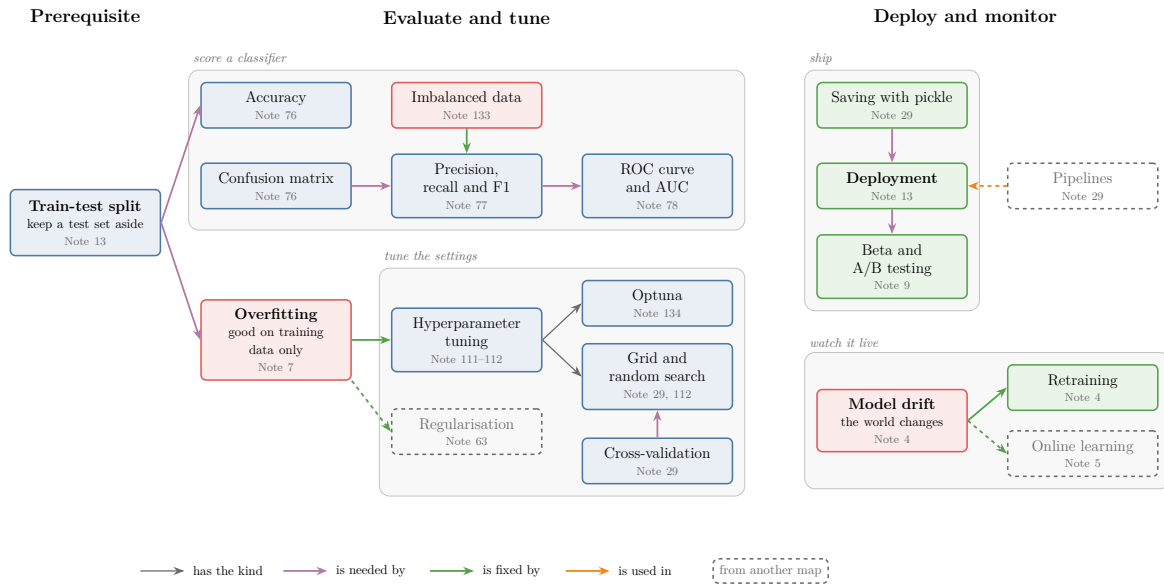


Figure 7: Concept map: Evaluating, tuning and production

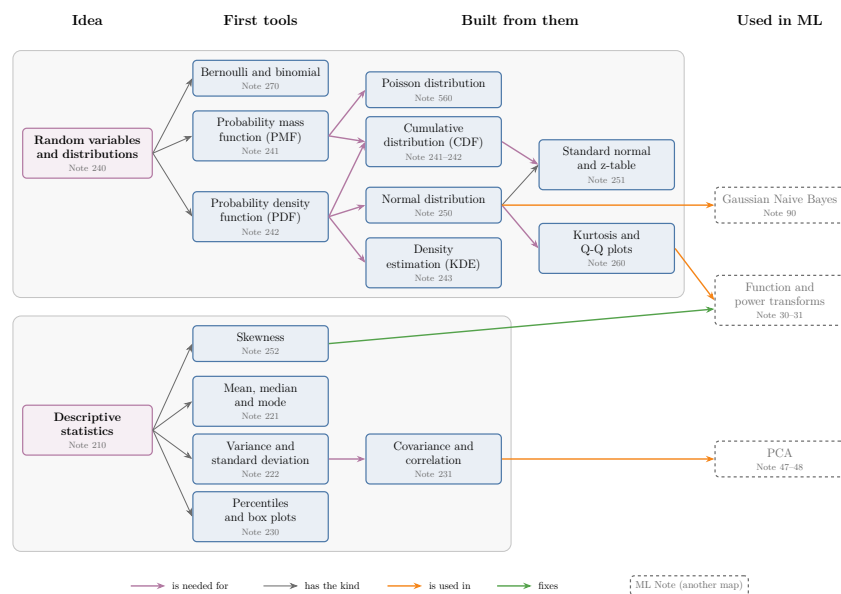


Figure 8: Concept map: Descriptive statistics and distributions

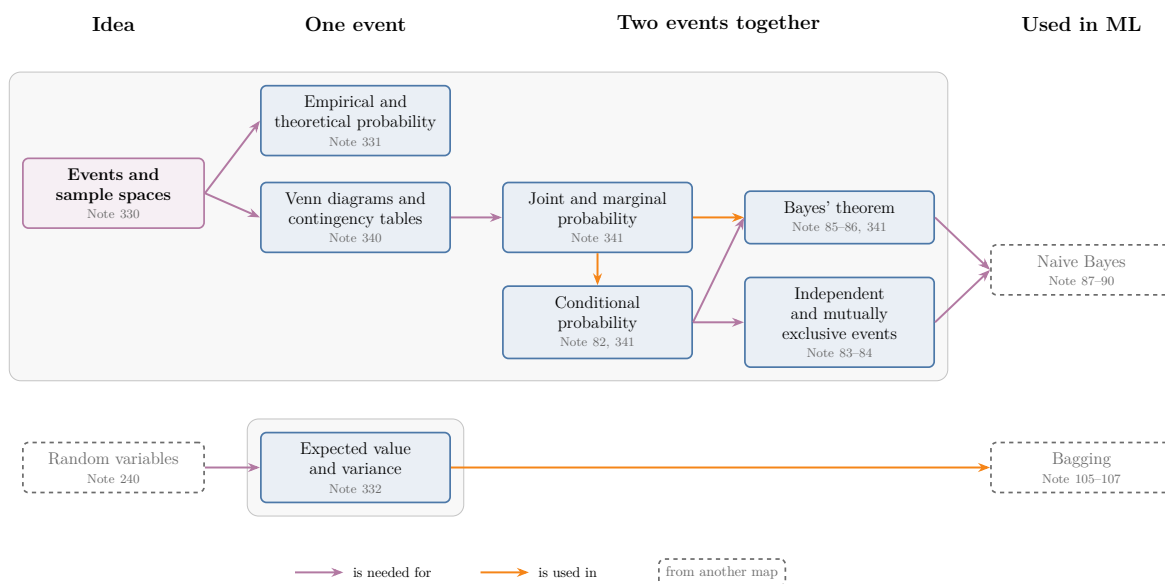


Figure 9: Concept map: Probability

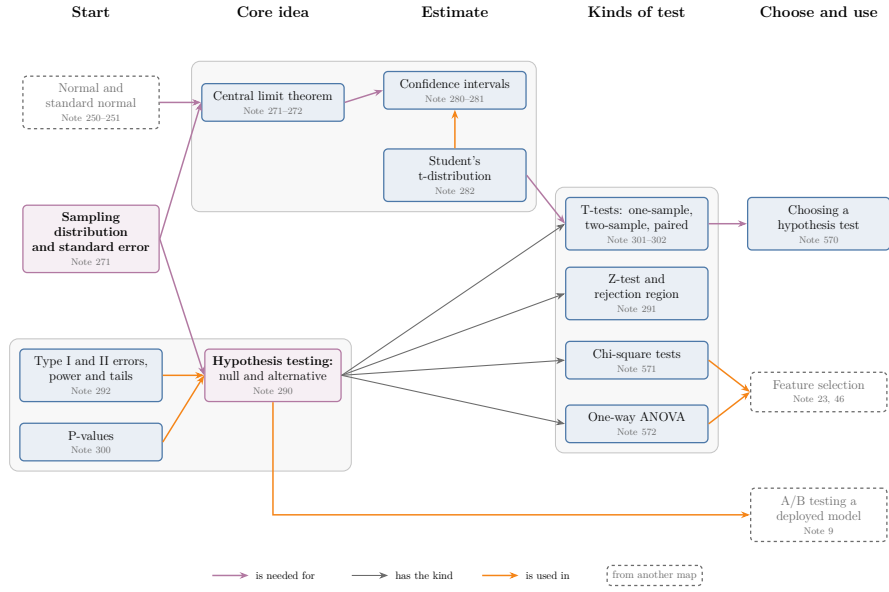


Figure 10: Concept map: Sampling, inference and hypothesis tests

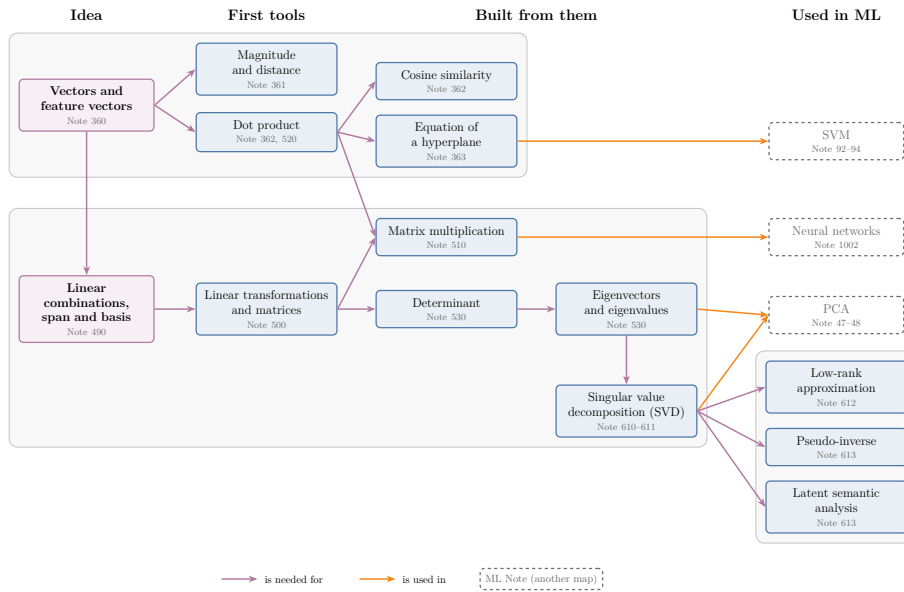


Figure 11: Concept map: Linear algebra

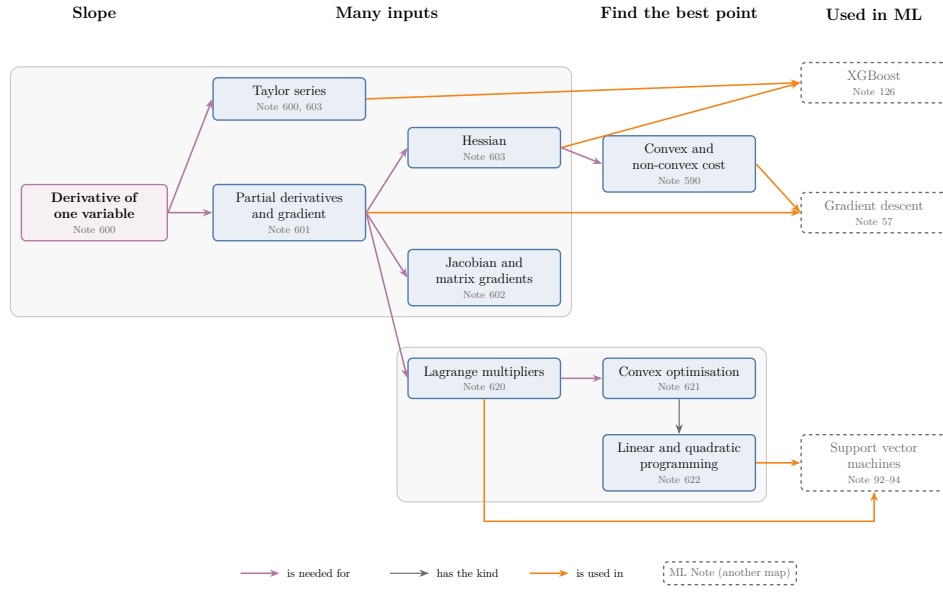


Figure 12: Concept map: Calculus and optimisation

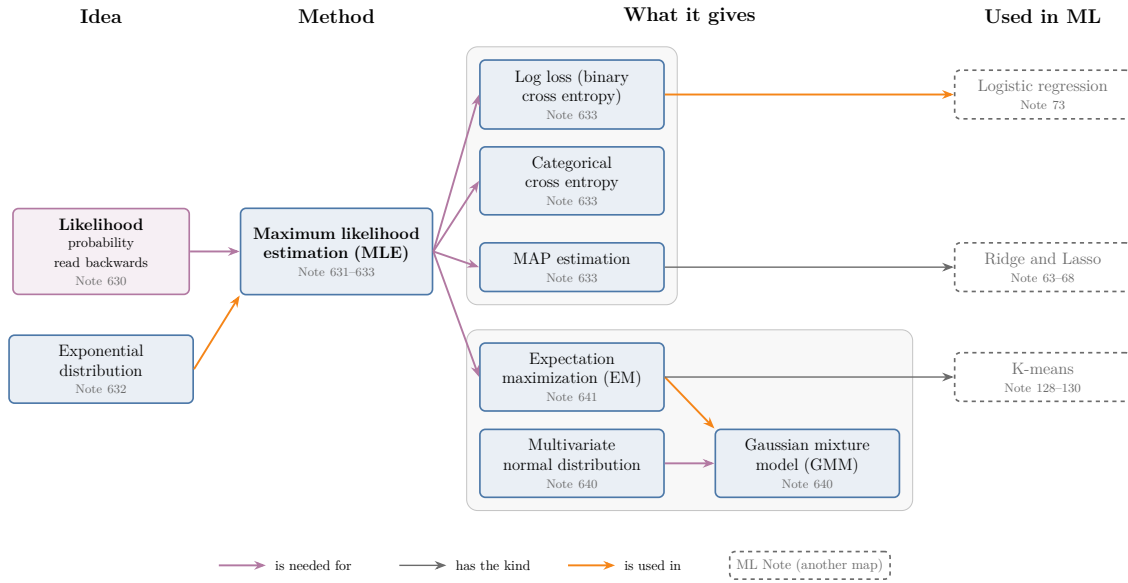


Figure 13: Concept map: Likelihood, MLE and mixture models

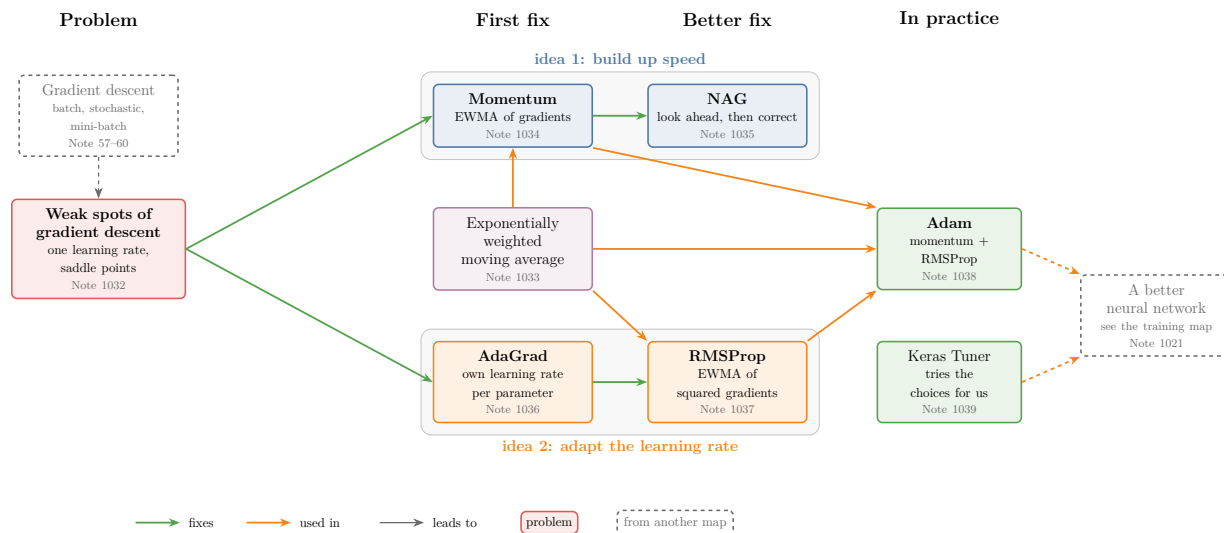


Figure 16: Concept map: Deep learning: optimizers

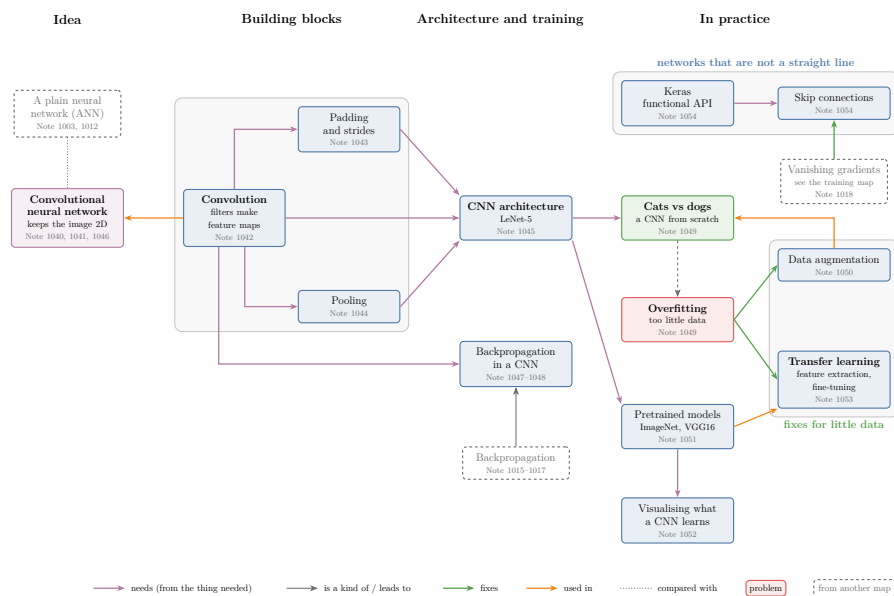


Figure 17: Concept map: Deep learning: convolutional networks

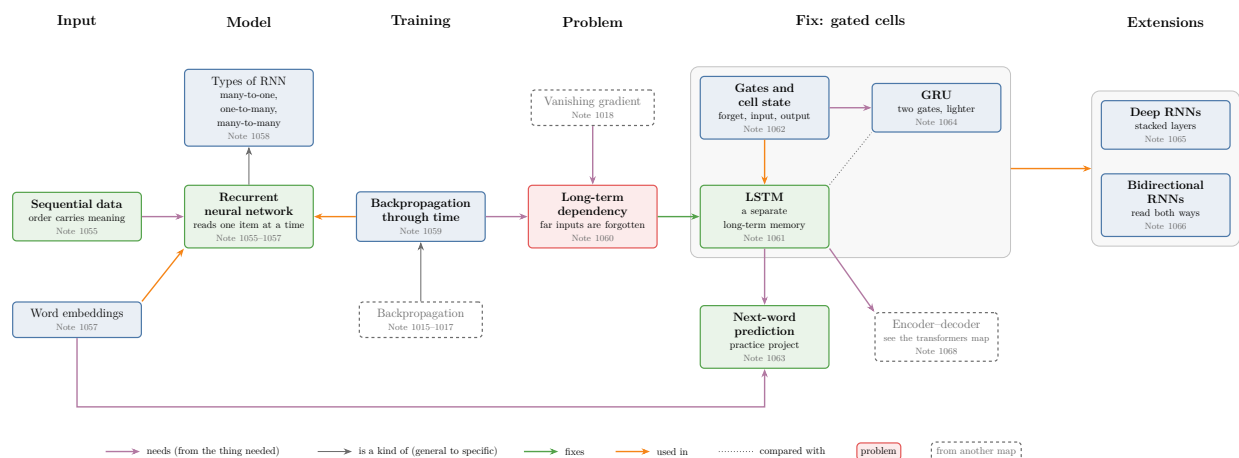


Figure 18: Concept map: Deep learning: recurrent networks

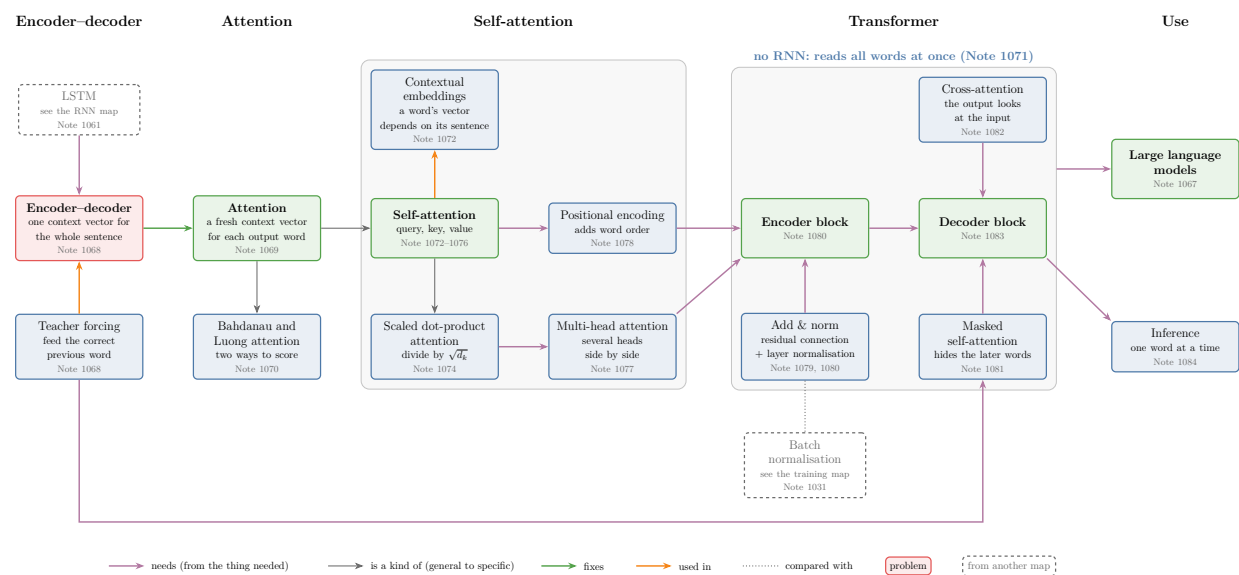


Figure 19: Concept map: Deep learning: LLMs, attention and transformers

4. The Learning path

Key point: Before each Note, read the Notes it builds on.

A Note is easiest to read when the ideas it uses are already familiar. The Notes that teach those ideas are its **prerequisites**: the Notes to read first. Each prerequisite has prerequisites of its own, so the reading order grows backwards from the Note we want, one round at a time.

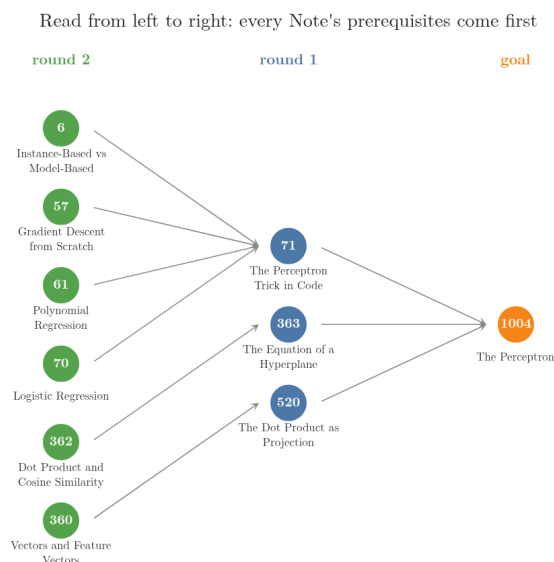


Figure 20: The reading order for the perceptron Note (1004), grown backwards. Round 1 adds the three Notes it builds on; round 2 adds the Notes that those build on. Arrows point from a prerequisite to the Note that needs it.

Figure 20 builds the reading order for the perceptron Note (Note 1004) step by step:

1. **Goal.** We want to read Note 1004.
2. **Round 1.** Its row in the table below lists Notes 71, 363 and 520: the perceptron trick in code, the equation of a hyperplane and the dot product.
3. **Round 2.** Each of those three has its own row. Note 71 builds on Notes 6, 57, 61 and 70; Note 363 on Note 362; Note 520 on Note 360.
4. **Reading.** Read the picture from left to right: green Notes first, then blue, then the goal. Every arrow points from a Note to a Note that needs it.

Each row of the table lists a Note's Concepts and the Notes to read first, with at most the four most recent. A row comes from the **needs**, **is a kind of**, **fixes** and **used in** Links of section 3: when a Concept of the Note builds on another Concept, the latest earlier Note that teaches that other Concept is read first. Notes marked *coming* or *deferred* are not written yet.

No.	Concepts	Read first	Note
1	Data mining, Machine learning	Note 2, Note 11, Note 12, Note 440	written

No.	Concepts	Read first	Note
2	Artificial intelligence, Deep learning, Features, Machine learning, Neural networks, Symbolic AI and expert systems	Note 12 , Note 440 , Note 500 , Note 510	written
3	Anomaly detection, Association rule learning, Classification problems, Clustering, Dimensionality reduction, Feature construction and splitting, Labelled data, PCA, Regression problems, Reinforcement learning, Semi-supervised learning, Supervised learning, Unsupervised learning	Note 132 , Note 500 , Note 530 , Note 610	written
4	Batch (offline) learning, Deployment, Model drift, Retraining	Note 7 , Note 9 , Note 13	written
5	Online learning, Out-of-core learning, Retraining, Stochastic gradient descent	Note 4 , Note 57 , Note 60	written
6	Feature scaling, Instance-based learning, K-nearest neighbours, Model-based learning	Note 23 , Note 29 , Note 46 , Note 361	written

No.	Concepts	Read first	Note
7	APIs, Deployment, Enough data, Feature construction and splitting, Feature engineering, Labelled data, MLOps and cost, Missing values, Outliers, Overfitting, Poor-quality data, Sampling noise and bias, Software integration, Underfitting, Web scraping	Note 16 , Note 23 , Note 29 , Note 41	written
8	Applications of ML, Data mining	Note 1 , Note 3	written
9	APIs, Beta and A/B testing, Bivariate and multivariate analysis, CSV files, Deployment, Ensemble learning, Exploratory data analysis, Feature construction and splitting, Feature engineering, Feature scaling, Feature selection, Framing an ML problem, Hyperparameter tuning, Imbalanced data, ML development life cycle, MLOps and cost, Missing values, Model drift, Outliers, Poor-quality data, Retraining, Saving models with pickle, Standardization, Univariate analysis, Web scraping	Note 99 , Note 290 , Note 571 , Note 572	written

No.	Concepts	Read first	Note
11	Features, One-hot encoding, Tensors	Note 23 , Note 27	written
12	Setup: conda, Jupyter and Colab	nothing	written
13	Accuracy, CSV files, Data leakage, Deployment, Exploratory data analysis, Feature scaling, Feature selection, Logistic regression, ML development life cycle, ML pipelines, Saving models with pickle, Standardization, Train-test split	Note 571 , Note 572 , Note 621 , Note 1004	written
14	Framing an ML problem	Note 3 , Note 4 , Note 5	written
15	CSV files	nothing	written
16	JSON and SQL data	nothing	written
17	APIs	Note 16	written
18	Web scraping	nothing	written
19	Correlation, Descriptive statistics, Exploratory data analysis, Variance	Note 9 , Note 22 , Note 222 , Note 223	written
20	Kernel density estimation (KDE), Outliers, Probability density function (PDF), Skewness, Univariate analysis	Note 19 , Note 41 , Note 210 , Note 243	written
21	Bivariate and multivariate analysis, Correlation	Note 9 , Note 19	written
22	Kurtosis and moments, Pandas Profiling	Note 12	written

No.	Concepts	Read first	Note
23	Binning and binarization, Encoding categorical data, Feature construction and splitting, Feature engineering, Feature extraction, Feature scaling, Feature selection, Feature transformation, Missing values, One-hot encoding, Outliers, Simple imputation (mean, median, mode, constant)	Note 67 , Note 99 , Note 571 , Note 572	written
24	Feature scaling, Standardization	Note 7 , Note 13 , Note 19 , Note 23	written
25	Normalization	Note 6	written
26	Encoding categorical data, Ordinal and label encoding	Note 13 , Note 23	written
27	Multicollinearity, One-hot encoding	Note 23	written
28	Column transformer	Note 23 , Note 30	written
29	Chi-square tests, Cross-validation, Deployment, Grid and random search, ML pipelines, Saving models with pickle	Note 23 , Note 28 , Note 290 , Note 341	written
30	Cross-validation, Function transformer, Q-Q plot	Note 22 , Note 23 , Note 240 , Note 242	written
31	Power transformer	Note 20 , Note 23 , Note 30	written
32	Binning and binarization, K-means	Note 7 , Note 23 , Note 128 , Note 641	written
33	Mixed variables	Note 3 , Note 7	written
34	Date and time features	Note 7 , Note 9	written

No.	Concepts	Read first	Note
35	Complete case analysis, Missing values	Note 7 , Note 9	written
36	Missing values, Simple imputation (mean, median, mode, constant)	Note 7 , Note 13	written
37	Missing values, Simple imputation (mean, median, mode, constant)	Note 7 , Note 13	written
38	Grid and random search, ML pipelines, Missing indicator, Missing values, Random sample imputation	Note 13 , Note 23 , Note 28 , Note 29	written
39	KNN imputer, Vector magnitude, distance and scalar operations	Note 6 , Note 7 , Note 38 , Note 360	written
40	Iterative imputation (MICE)	Note 7 , Note 23 , Note 50	written
41	Capping (winsorization), IQR outlier method, Outliers, Percentile outlier method, Trimming outliers, Z-score outlier method	Note 19 , Note 20 , Note 230 , Note 240	written
42	Capping (winsorization), Normal distribution, Trimming outliers, Z-score outlier method	Note 19 , Note 20 , Note 41 , Note 210	written
43	Capping (winsorization), IQR outlier method, Trimming outliers	Note 7 , Note 9 , Note 19 , Note 41	written

No.	Concepts	Read first	Note
44	Capping (winsorization), Percentile outlier method, Trimming outliers	Note 7 , Note 230	written
45	Feature construction and splitting	Note 7 , Note 23 , Note 29	written
46	Curse of dimensionality, Dimensionality reduction, Feature extraction, Feature selection	Note 67 , Note 99 , Note 571 , Note 572	written
47	Feature extraction, PCA, Variance	Note 222 , Note 500 , Note 530 , Note 610	written
48	Covariance and covariance matrix, Dot product, Eigenvectors and eigenvalues, Linear transformations and matrices, PCA	Note 360 , Note 490 , Note 530 , Note 610	written
49	PCA	Note 48 , Note 500 , Note 530 , Note 610	written
50	Best-fit line and squared error, Simple linear regression	Note 51 , Note 52 , Note 56 , Note 57	written
51	Best-fit line and squared error, Derivatives of one variable, Ordinary least squares (closed form), Simple linear regression	Note 48 , Note 52 , Note 56 , Note 57	written
52	Regression metrics	Note 50	written
53	Equation of a hyperplane, Multiple linear regression	Note 50 , Note 54 , Note 58 , Note 362	written
54	Multiple linear regression, Normal equation	Note 51 , Note 58 , Note 490 , Note 602	written

No.	Concepts	Read first	Note
55	Multiple linear regression, Normal equation	Note 51 , Note 58 , Note 490 , Note 602	written
56	Assumptions of linear regression	Note 27 , Note 30	written
57	Convex and non-convex loss, Gradient descent, Learning rate, Partial derivatives and gradients	Note 6 , Note 50 , Note 600 , Note 603	written
58	Batch gradient descent	Note 57	written
59	Stochastic gradient descent	Note 57	written
60	Mini-batch gradient descent	Note 57	written
61	Overfitting, Polynomial features, Polynomial regression, Underfitting	Note 13 , Note 53	written
62	Bias-variance trade-off	Note 7 , Note 61	written
63	Regularisation, Ridge regression	Note 62 , Note 361 , Note 620 , Note 633	written
64	Ridge regression	Note 62 , Note 63 , Note 620 , Note 633	written
65	Ridge regression	Note 62 , Note 63 , Note 620 , Note 633	written
66	Ridge regression	Note 62 , Note 63 , Note 620 , Note 633	written
67	Lasso regression	Note 63 , Note 620 , Note 633	written
68	Lasso regression	Note 63 , Note 620 , Note 633	written
69	Elastic Net	Note 27 , Note 63 , Note 67	written
70	Equation of a hyperplane, Logistic regression, Perceptron trick	Note 73 , Note 362 , Note 621 , Note 1004	written

No.	Concepts	Read first	Note
71	Logistic regression, Perceptron trick	Note 73 , Note 363 , Note 621 , Note 1004	written
72	Logistic regression, Sigmoid function	Note 73 , Note 621 , Note 1004 , Note 1027	written
73	Log loss (binary cross entropy), Logistic regression	Note 621 , Note 631 , Note 1004 , Note 1014	written
74	Sigmoid function	Note 1027	written
75	Logistic regression	Note 72 , Note 73 , Note 621 , Note 1004	written
76	Accuracy, Confusion matrix	Note 9 , Note 13	written
77	Precision, recall and F1	Note 9 , Note 76	written
78	ROC curve and AUC	Note 76 , Note 77	written
79	Softmax regression	Note 11 , Note 13	written
80	Polynomial features	nothing	written
81	Hyperparameter tuning	Note 7	written
82	Conditional probability	Note 330 , Note 341	written
83	Independent and mutually exclusive events	Note 341	written
84	Independent and mutually exclusive events	Note 341	written
85	Bayes' theorem	Note 341	written
86	Bayes' theorem	Note 341	written
87	Naive Bayes	Note 20 , Note 83 , Note 240 , Note 341	written
88	Naive Bayes	Note 20 , Note 83 , Note 240 , Note 341	written
89	Naive Bayes	Note 20 , Note 83 , Note 240 , Note 341	written
90	Naive Bayes, Normal distribution, Probability density function (PDF)	Note 3 , Note 83 , Note 210 , Note 341	written

No.	Concepts	Read first	Note
91	Accuracy, Cross-validation, Curse of dimensionality, Data leakage, Decision surface and boundary, Grid and random search, Hyperparameter tuning, K-nearest neighbours, Overfitting, Underfitting	Note 6 , Note 13 , Note 23 , Note 361	written
92	Support vector machines	Note 362 , Note 363 , Note 620 , Note 622	written
93	Support vector machines	Note 362 , Note 363 , Note 620 , Note 622	written
94	Hinge loss and soft margin, Support vector machines	Note 362 , Note 363 , Note 620 , Note 622	written
95	Kernel trick	nothing	written
96	Kernel trick	nothing	written
97	Decision trees, Entropy, information gain and Gini	Note 6 , Note 91	written
98	Decision trees, Hyperparameter tuning	Note 6 , Note 7 , Note 91 , Note 97	written
99	Feature importance, Grid and random search, Regression trees	Note 23 , Note 29 , Note 52 , Note 97	written
100	Decision trees, Feature importance	Note 6 , Note 91 , Note 97	written
101	Boosting, Ensemble learning	Note 7 , Note 62 , Note 97	written
102	Bernoulli and binomial distributions, Voting ensembles	Note 9 , Note 29 , Note 83 , Note 210	written
103	Cross-validation, Voting ensembles	Note 9 , Note 83 , Note 241	written

No.	Concepts	Read first	Note
104	Cross-validation, Voting ensembles	Note 9 , Note 83 , Note 241	written
105	Bagging, OOB score	Note 29 , Note 62 , Note 97 , Note 332	written
106	Bagging, Grid and random search, OOB score	Note 29 , Note 62 , Note 97 , Note 332	written
107	Bagging, Grid and random search, OOB score	Note 29 , Note 62 , Note 97 , Note 332	written
108	Random forest	Note 29 , Note 97 , Note 99 , Note 105	written
109	Bias-variance trade-off, Random forest	Note 61 , Note 97 , Note 99 , Note 105	written
110	Random forest	Note 29 , Note 97 , Note 99 , Note 105	written
111	Hyperparameter tuning, Random forest	Note 29 , Note 97 , Note 99 , Note 105	written
112	Cross-validation, Grid and random search, Random forest	Note 23 , Note 97 , Note 99 , Note 105	written
113	OOB score, Random forest	Note 29 , Note 97 , Note 99 , Note 105	written
114	Feature importance, Permutation importance, Random forest	Note 9 , Note 29 , Note 97 , Note 105	written
115	AdaBoost	Note 9 , Note 97 , Note 101	written
116	AdaBoost	Note 9 , Note 97 , Note 101	written
117	AdaBoost	Note 9 , Note 97 , Note 101	written
118	AdaBoost, Cross-validation, Grid and random search, Hyperparameter tuning, Learning rate	Note 13 , Note 23 , Note 97 , Note 101	written

No.	Concepts	Read first	Note
119	Boosting	Note 7 , Note 9 , Note 62	written
120	Gradient boosting, Learning rate	Note 72 , Note 73 , Note 99 , Note 101	written
121	Gradient boosting	Note 72 , Note 73 , Note 99 , Note 101	written
122	Gradient boosting	Note 72 , Note 73 , Note 99 , Note 101	written
123	Missing values, XGBoost	Note 63 , Note 120 , Note 600 , Note 603	written
124	XGBoost	Note 63 , Note 120 , Note 600 , Note 603	written
125	XGBoost	Note 63 , Note 120 , Note 600 , Note 603	written
126	Hessian and multivariate Taylor, Taylor series, XGBoost	Note 120 , Note 530 , Note 600 , Note 601	written
127	Cross-validation, Stacking and blending	Note 9	written
128	Clustering, Elbow method and WCSS, K-means	Note 3 , Note 6 , Note 641	written
129	Clustering, Elbow method and WCSS, K-means	Note 3 , Note 6 , Note 641	written
130	Clustering, K-means	Note 3 , Note 6 , Note 128 , Note 641	written
131	Clustering, Hierarchical clustering	Note 3	written
132	Anomaly detection, Clustering, DBSCAN	Note 3	written
133	Balanced random forest, Cost-sensitive learning, Imbalanced data, Random under- and oversampling, SMOTE	Note 29 , Note 73 , Note 78 , Note 108	written
134	Bayesian optimisation, Optuna	Note 9 , Note 29	written

No.	Concepts	Read first	Note
210	Descriptive statistics, Inferential statistics, Probability distributions	Note 220 , Note 223 , Note 240 , Note 280	written
220	Discrete and continuous data, Inferential statistics, Population, sample, parameter and statistic	Note 7 , Note 280	written
221	Descriptive statistics, Measures of central tendency	Note 9 , Note 223	written
222	Bessel's correction, Descriptive statistics, Variance	Note 9 , Note 220 , Note 223	written
223	Descriptive statistics, Frequency tables	Note 9	written
230	Descriptive statistics, Percentiles, quartiles and box plots	Note 9 , Note 223	written
231	Correlation, Correlation and causation, Covariance and covariance matrix	Note 19	written
240	Normal distribution, Probability distributions, Random variables	Note 20 , Note 220	written
241	Bernoulli and binomial distributions, Cumulative distribution function (CDF), Probability mass function (PMF), Uniform distribution	Note 20 , Note 210	written

No.	Concepts	Read first	Note
242	Cumulative distribution function (CDF), Log-normal distribution, Poisson distribution, Probability density function (PDF)	Note 210 , Note 241 , Note 332	written
243	Density estimation, Kernel density estimation (KDE)	Note 20 , Note 220	written
250	Normal distribution	Note 20 , Note 210	written
251	Standard normal and the z-table, Z-score outlier method	Note 19 , Note 20 , Note 240 , Note 241	written
252	Skewness	Note 19	written
253	Cumulative distribution function (CDF), Kernel density estimation (KDE)	Note 20 , Note 241 , Note 243	written
260	Kurtosis and moments, Q-Q plot	Note 240	written
261	Log-normal distribution, Uniform distribution	Note 210	written
262	Pareto distribution and power laws	nothing	written
270	Bernoulli and binomial distributions	Note 210	written
271	Central limit theorem, Sampling distribution and standard error	Note 220 , Note 240	written
272	Central limit theorem	Note 240 , Note 271	written
280	Confidence intervals	Note 251 , Note 271 , Note 282	written
281	Confidence intervals	Note 251 , Note 271 , Note 282	written
282	Confidence intervals, Student's t-distribution	Note 222 , Note 251 , Note 271	written

No.	Concepts	Read first	Note
290	Hypothesis testing: null and alternative	Note 210 , Note 271 , Note 292 , Note 300	written
291	Hypothesis testing: null and alternative, Z-test and rejection regions	Note 251 , Note 271 , Note 292 , Note 300	written
292	Beta and A/B testing, Type I and II errors, power, tails	Note 4 , Note 290	written
300	P-values	Note 241	written
301	T-tests: one-sample, two-sample, paired	Note 282 , Note 290	written
302	T-tests: one-sample, two-sample, paired	Note 282 , Note 290	written
330	Events and sample spaces	nothing	written
331	Empirical vs theoretical probability, probability rules	Note 330	written
332	Expected value and variance of a random variable	Note 240	written
340	Empirical vs theoretical probability, probability rules, Venn diagrams and contingency tables	Note 330	written
341	Bayes' theorem, Conditional probability, Joint and marginal probability	Note 330 , Note 340	written
350	Linear algebra roadmap	nothing	written
360	Bag of words, Vectors and feature vectors	Note 11	written
361	Vector magnitude, distance and scalar operations	Note 360	written
362	Cosine similarity, Dot product	Note 360	written

No.	Concepts	Read first	Note
363	Equation of a hyperplane	Note 362	written
440	Role of mathematics in ML	nothing	written
490	Linear combinations, span and basis	Note 360	written
500	Linear transformations and matrices	Note 490	written
510	Matrix multiplication as composition	Note 362 , Note 500	written
520	Dot product	Note 360	written
530	Determinant, Eigenvectors and eigenvalues	Note 500	written
560	Poisson distribution	Note 241 , Note 332	written
570	Choosing a hypothesis test, Correlation significance test, One-sample proportion test, T-tests: one-sample, two-sample, paired	Note 19 , Note 282 , Note 290 , Note 291	written
571	Chi-square tests	Note 290 , Note 341	written
572	One-way ANOVA	Note 19 , Note 290	written
580	How to learn the maths for ML	Note 440	written
590	Convex and non-convex loss	Note 603	written
600	Derivatives of one variable, Taylor series	nothing	written
601	Partial derivatives and gradients	Note 600	written
602	Jacobian and matrix gradients	Note 500 , Note 510 , Note 601	written
603	Hessian and multivariate Taylor, Taylor series	Note 530 , Note 600 , Note 601	written
610	Singular value decomposition	Note 500 , Note 530	written

No.	Concepts	Read first	Note
611	Singular value decomposition	Note 500 , Note 530	written
612	Low-rank approximation (truncated SVD)	Note 610	written
613	Latent semantic analysis, Moore-Penrose pseudo-inverse	Note 27 , Note 360 , Note 610	written
620	Lagrange multipliers, KKT and duality	Note 601	written
621	Convex sets and convex optimisation	Note 57 , Note 620	written
622	Linear and quadratic programming	Note 621	written
630	Likelihood, Normal distribution	Note 20 , Note 210	written
631	Maximum likelihood estimation (MLE)	Note 600 , Note 601 , Note 630 , Note 632	written
632	Exponential distribution, Maximum likelihood estimation (MLE)	Note 20 , Note 600 , Note 601 , Note 630	written
633	Categorical and sparse categorical cross-entropy, Log loss (binary cross entropy), MAP estimation, Maximum likelihood estimation (MLE)	Note 601 , Note 630 , Note 632 , Note 1014	written
640	Gaussian mixture model (GMM), Multivariate normal distribution	Note 240 , Note 341 , Note 631 , Note 641	written
641	Expectation maximization (EM), Gaussian mixture model (GMM), K-means	Note 128 , Note 341 , Note 631 , Note 640	written
1001	What deep learning is	Note 1	written

No.	Concepts	Read first	Note
1002	Neural networks, Representation learning, What deep learning is	Note 1 , Note 500 , Note 510	written
1003	History of deep learning, Multi-layer perceptron (MLP), Types of neural networks, Universal approximation theorem	Note 1007 , Note 1008 , Note 1015 , Note 1027	written
1004	Perceptron	Note 70 , Note 362 , Note 363	written
1005	Perceptron trick	Note 363	written
1006	Perceptron loss	Note 5 , Note 70	written
1007	Problem with the perceptron (XOR)	Note 1004	written
1008	MLP notation and parameter count	nothing	written
1009	Multi-layer perceptron (MLP), Universal approximation theorem	Note 1007 , Note 1008 , Note 1015 , Note 1027	written
1010	Forward propagation	Note 510 , Note 1008	written
1011	ANN for classification, Keras workflow, Training curves (History)	Note 79 , Note 633 , Note 1003 , Note 1015	written
1012	ANN for classification, Categorical and sparse categorical cross-entropy, Keras workflow, Training curves (History)	Note 631 , Note 1003 , Note 1014 , Note 1015	written
1013	ANN for regression, Keras workflow, Training curves (History)	Note 25 , Note 52 , Note 1003 , Note 1015	written

No.	Concepts	Read first	Note
1014	Categorical and sparse categorical cross-entropy, Huber loss, Log loss (binary cross entropy), Loss functions in deep learning	Note 11 , Note 79 , Note 631 , Note 1010	written
1015	Backpropagation	Note 1010 , Note 1014 , Note 1019 , Note 1020	written
1016	Backpropagation	Note 1010 , Note 1014 , Note 1019 , Note 1020	written
1017	Backpropagation, Convex and non-convex loss, Gradient descent, Learning rate	Note 1010 , Note 1014 , Note 1019 , Note 1020	written
1018	Exploding gradient and gradient clipping, ReLU, Sigmoid function, Vanishing gradient	Note 1011 , Note 1015 , Note 1027	written
1019	Backpropagation, Memoization	Note 602 , Note 1010 , Note 1014 , Note 1020	written
1020	Batch gradient descent, Batch size in Keras, Gradient descent, Mini-batch gradient descent, Stochastic gradient descent	Note 50 , Note 57 , Note 600 , Note 601	written
1021	Early stopping, Hyperparameter tuning, Improving a neural network	Note 1029 , Note 1031 , Note 1032 , Note 1039	written
1022	Early stopping, Keras workflow, Training curves (History)	Note 7 , Note 1003	written

No.	Concepts	Read first	Note
1023	Feature scaling, Scaling inputs for neural networks, Standardization	Note 19 , Note 23 , Note 25 , Note 57	written
1024	Dropout	Note 7 , Note 63	written
1025	Dropout, Keras workflow	Note 7 , Note 63 , Note 1003	written
1026	Keras workflow, L1 and L2 regularisation in neural networks, Overfitting, Regularisation	Note 13 , Note 57 , Note 361 , Note 1003	written
1027	Activation functions, ReLU, Sigmoid function, Tanh	Note 1004 , Note 1018	written
1028	Dying ReLU problem, Leaky ReLU, PReLU, ELU and SELU, ReLU	Note 1018 , Note 1027	written
1029	Exploding gradient and gradient clipping, Vanishing gradient, Weight initialisation	Note 72 , Note 1011 , Note 1015	written
1030	Vanishing gradient, Weight initialisation, Xavier and He initialisation	Note 1011 , Note 1015 , Note 1018 , Note 1027	written
1031	Batch normalisation, Covariate shift, Standardization	Note 13 , Note 19 , Note 60 , Note 1033	written
1032	Local minima and saddle points, Optimizers in deep learning	Note 57 , Note 60 , Note 621 , Note 1015	written
1033	Exponentially weighted moving average (EWMA)	nothing	written
1034	Local minima and saddle points, SGD with momentum	Note 621 , Note 1032 , Note 1033	written
1035	Nesterov accelerated gradient (NAG)	Note 1032 , Note 1034	written
1036	AdaGrad	Note 57 , Note 1032	written

No.	Concepts	Read first	Note
1037	RMSProp	Note 1032 , Note 1033 , Note 1036	written
1038	Adam	Note 1032 , Note 1033 , Note 1034 , Note 1037	written
1039	Hyperparameter tuning, Keras Tuner	Note 7 , Note 1011	written
1040	Convolutional neural network (CNN)	Note 1002 , Note 1042 , Note 1044	written
1041	Convolutional neural network (CNN)	Note 1002 , Note 1042 , Note 1044	written
1042	Convolution operation and feature maps	nothing	written
1043	Padding and strides	Note 1042	written
1044	Pooling	Note 1042	written
1045	CNN architecture (LeNet-5)	Note 1042 , Note 1043 , Note 1044	written
1046	Convolutional neural network (CNN)	Note 1002 , Note 1042 , Note 1044	written
1047	Backpropagation in a CNN	Note 1015 , Note 1042	written
1048	Backpropagation in a CNN	Note 1015 , Note 1042	written
1049	Image classification with a CNN (cats vs dogs)	Note 1011 , Note 1023 , Note 1045 , Note 1050	written
1050	Data augmentation	Note 7 , Note 63	written
1051	Pretrained models and ImageNet	Note 3 , Note 1045	written
1052	Visualising what a CNN learns	Note 1042 , Note 1051	written
1053	Transfer learning (feature extraction and fine-tuning)	Note 7 , Note 1051	written
1054	Keras functional API, Skip connections	Note 1011 , Note 1018	written
1055	Recurrent neural network (RNN), Sequence padding, Sequential data	Note 1027 , Note 1056 , Note 1057 , Note 1059	written

No.	Concepts	Read first	Note
1056	Parameter sharing across time steps, Recurrent neural network (RNN)	Note 1027 , Note 1055 , Note 1057 , Note 1059	written
1057	Recurrent neural network (RNN), Sequence padding, Tokenization and integer encoding of text, Word embeddings	Note 1027 , Note 1055 , Note 1056 , Note 1059	written
1058	Sequence-to-sequence (encoder-decoder), Types of RNN (many-to-one, one-to-many, many-to-many)	Note 1055 , Note 1061 , Note 1068	written
1059	Backpropagation through time (BPTT), Parameter sharing across time steps	Note 1015	written
1060	Exploding gradient and gradient clipping, Long-term dependency problem, Vanishing gradient	Note 72 , Note 1011 , Note 1015 , Note 1059	written
1061	LSTM (long short-term memory)	Note 1055 , Note 1060 , Note 1062	written
1062	LSTM (long short-term memory), LSTM gates (forget, input, output) and cell state	Note 1055 , Note 1060	written
1063	LSTM (long short-term memory), Next-word prediction with an LSTM	Note 1055 , Note 1057 , Note 1060 , Note 1062	written
1064	GRU (gated recurrent unit)	Note 1055 , Note 1062	written
1065	Deep (stacked) RNNs	Note 1055 , Note 1061 , Note 1064	written

No.	Concepts	Read first	Note
1066	Bidirectional RNNs	Note 1055 , Note 1061 , Note 1064	written
1067	Large language models (LLMs)	Note 1071 , Note 1087	written
1068	Sequence-to-sequence (encoder-decoder), Teacher forcing	Note 1058 , Note 1061	written
1069	Attention mechanism, Bahdanau (additive) attention	Note 1058	written
1070	Bahdanau (additive) attention, Luong (multiplicative) attention	Note 1069	written
1071	Transformer	Note 1058 , Note 1080 , Note 1083 , Note 1085	written
1072	Contextual embeddings, Self-attention (query, key, value)	Note 1057 , Note 1069 , Note 1086	written
1073	Self-attention (query, key, value)	Note 1069	written
1074	Scaled dot-product attention	Note 1072	written
1075	Self-attention (query, key, value)	Note 1069	written
1076	Self-attention (query, key, value)	Note 1069	written
1077	Multi-head attention	Note 1074	written
1078	Positional encoding	Note 1072	written
1079	Layer normalisation	nothing	written
1080	Residual connections and add & norm, Transformer, Transformer encoder	Note 1079 , Note 1083 , Note 1085 , Note 1089	written
1081	Masked self-attention, Teacher forcing	Note 1072	written
1082	Cross-attention	nothing	written

No.	Concepts	Read first	Note
1083	Transformer, Transformer decoder	Note 1080 , Note 1081 , Note 1082 , Note 1085	written
1084	Transformer inference (autoregressive decoding, KV cache, beam search)	Note 1083	written
1085	Label smoothing, Learning-rate warm-up schedule, The transformer end to end (capstone)	Note 1038 , Note 1080 , Note 1083 , Note 1084	written
1086	Meaning as direction in embedding space	Note 362 , Note 1057	written
1087	Decoder-only GPT, GELU activation	Note 1071 , Note 1081 , Note 1083 , Note 1088	written
1088	Logit lens, Unembedding, logits, temperature and sampling	Note 79	written
1089	MLP blocks as fact storage	Note 1087	written
1090	Superposition and nearly perpendicular directions	Note 1086	written

5. The Algorithm chooser

Key point: Two questions narrow the choice: do we have labelled outputs, and are they numbers or categories?

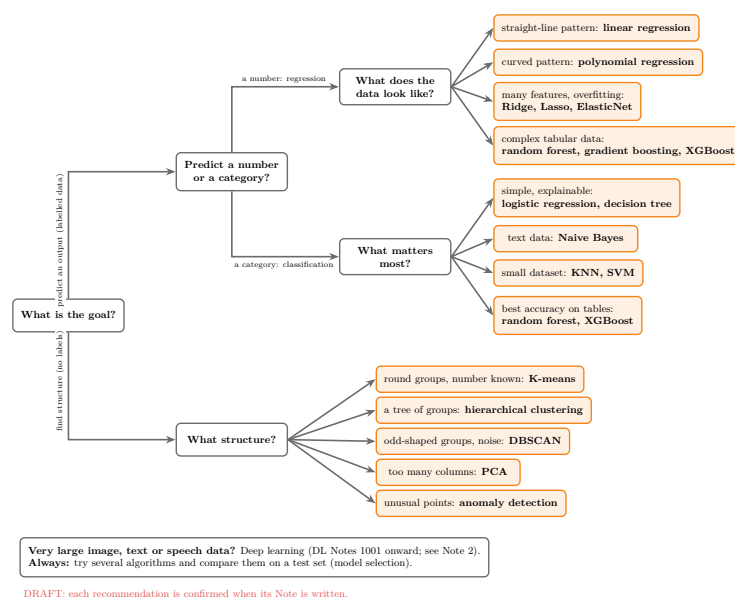


Figure 21: The Algorithm chooser (draft)

Figure 21 is a starting point, not a rule: in practice we try several suitable algorithms and compare them on a test set. It is a draft until the algorithm Notes are written.

6. Sources

Built from - CampusX, “100 Days of Machine Learning”, YouTube playlist (134 videos), https://www.youtube.com/playlist?list=PLKnIA16_Rmvbr7zKYQuBfsVkjoLcJgxHH - CampusX, “100 Days of Deep Learning”, YouTube playlist (84 videos), <https://www.youtube.com/playlist?list=P...> - CampusX, “Maths for Machine Learning”, YouTube playlist (23 sessions), <https://www.youtube.com/playlist?list=3vADST> - Every Note listed in the Learning path: each Note’s own Sources name the video, book or paper it was built from.

Other references - The Concept map data: `course_map/concepts.yaml` in this repository.

7. Key terms

Term	Meaning
Course map (G-2160)	The overview that ties all Notes together, in four views
Pipeline map (G-2161)	Where each Concept sits among the 14 steps of an ML project
Concept map (G-2162)	How Concepts are connected by Links
Learning path (G-2163)	Which Notes to read before which
Algorithm chooser (G-2164)	A flowchart from a problem’s properties to suitable algorithms

Term	Meaning
Concept (G-2165)	One idea on the Course map
Link (G-2166)	A labelled connection between two Concepts
Draft / confirmed (G-2167)	Guessed from titles / checked against a written Note